Report_DEB

1. Importation des données

```
file_path <- here::here("mod/D_DEB_SAAWYe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```r
NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 =
seeds <- c(C1 = 33113, C2 = 6666, C3 = 2212)

text_priors <- '
Distrib (Fm,           TruncNormal,           0.1,      0.2,  0.01,      0.8);
Distrib (pAm,          TruncNormal_cv,        100,      0.2,    40,    2000);
Distrib (r_pAm_pM,     TruncNormal_cv,        0.8,      0.1,    0.1,      2);
Distrib (v,            TruncNormal_cv,    0.18505,      0.1,  0.001,    2);
Distrib (kap,          Uniform,               0,     1);
Distrib (kapA,         Uniform,               0.1,     0.9);
Distrib (Ehp,          TruncNormal_cv,         100,      0.2,    40,    300);
Distrib (E_coc,        TruncNormal_cv,          30,      0.2,     1,    300);
Distrib (rOM_ClxHorse, LogUniform,         0.00001,      0.3);
Distrib (dv,           TruncNormal_cv,           2,      0.5,     0,     10);
Distrib (wE,           LogUniform,      0.000000001,     0.1);

# Calcul des vraissemblances
Distrib (Sigma_W, TruncNormal,  5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,       Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIte

# ./mcsim.DEBcali DEB_C1.in
```

`OM_horse` and `OM_soil` distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

```
                 Fm.1.   pAm.1. r_pAm_pM.1.       v.1.     kap.1.    kapA.1.
Result_mode 0.43022000 185.0380  0.80045100 0.20176100 0.92458200 0.71123300
```

```
2.5%        0.31910920 158.7960  0.65039880 0.15526600 0.88094800 0.60038500
97.5%       0.50967500 207.6530  0.88899200 0.21858760 0.94347600 0.78717300
Min         0.27075500  90.5855  0.56490300 0.13140100 0.43482600 0.19504900
Max         0.58464700 224.6830  0.97241100 0.25509500 0.96301800 0.85624800
Result_mean 0.40860761 184.9805  0.77249960 0.18858013 0.91602825 0.69432769
Result_sd   0.04976404  11.9681  0.06276992 0.01656593 0.01671322 0.04809283
               Ehp.1.  E_coc.1. rOM_ClxHorse.1.     dv.1.        wE.1.
Result_mode  93.95090 16.137100     0.003262230 3.9923000 1.006290e-05
2.5%         72.81794 11.694200     0.002033984 2.8651250 1.945200e-09
97.5%       141.79300 25.045300     0.005338830 4.7220230 1.070360e-05
Min          61.29110  8.543220     0.000125481 0.5220400 1.099830e-09
Max         166.03300 32.675200     0.187905000 5.4313600 1.139010e-05
Result_mean 106.67458 18.143717     0.003521053 3.7659873 7.122750e-06
Result_sd    16.85487  3.356669     0.001358310 0.4856263 3.564934e-06
            Sigma_W.1.
Result_mode 0.10594700
2.5%        0.10064400
97.5%       0.12879400
Min         0.09019160
Max         4.53346000
Result_mean 0.11399039
Result_sd   0.03977369

Number of observations, n =  277
Number of parameters, k =  11
Log-likelihood, LL =  13.51421
BIC =  34.83577


Potential scale reduction factors:

                Point est. Upper C.I.
dv.1.                 1.09       1.29
E_coc.1.              1.06       1.20
Ehp.1.                1.32       1.88
Fm.1.                 1.06       1.20
kap.1.                1.18       1.52
kapA.1.               1.08       1.23
pAm.1.                1.24       1.66
r_pAm_pM.1.           1.02       1.05
rOM_ClxHorse.1.       1.07       1.22
Sigma_W.1.            1.03       1.11
```

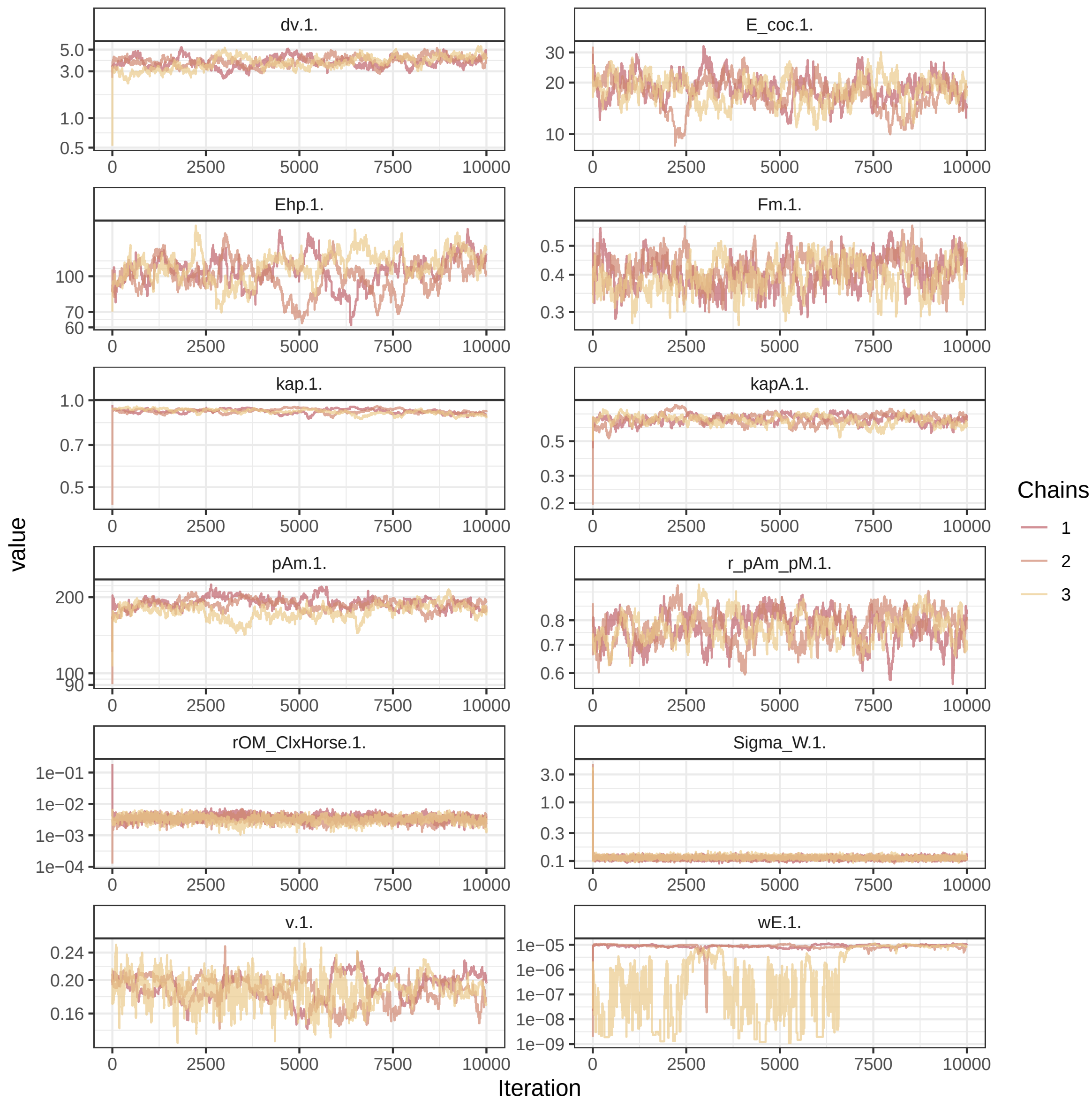


Figure 1: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

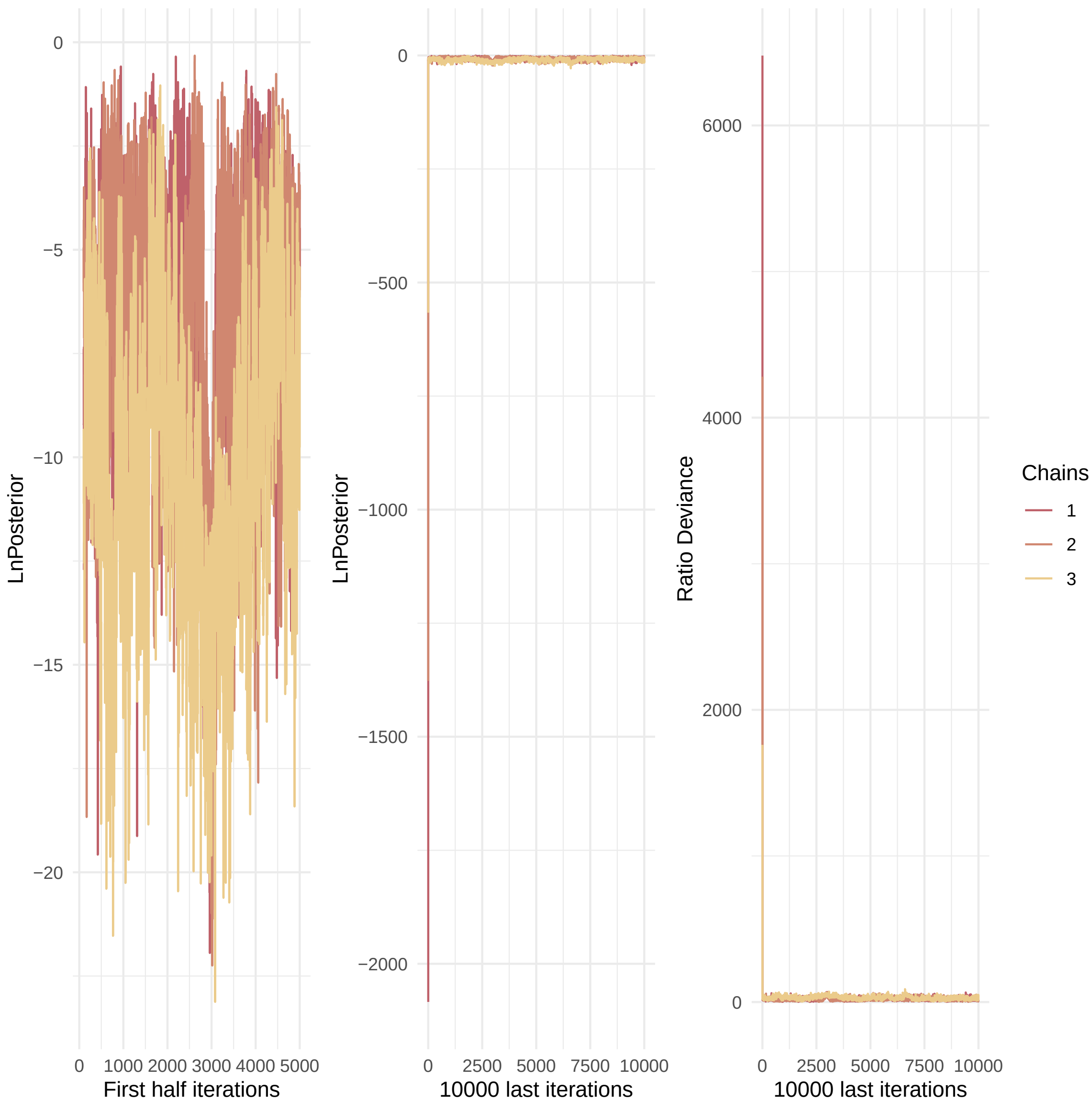


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

```
v.1.                 1.22       1.63
wE.1.                1.48       3.66

Multivariate psrf

1.53
```

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE])  # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

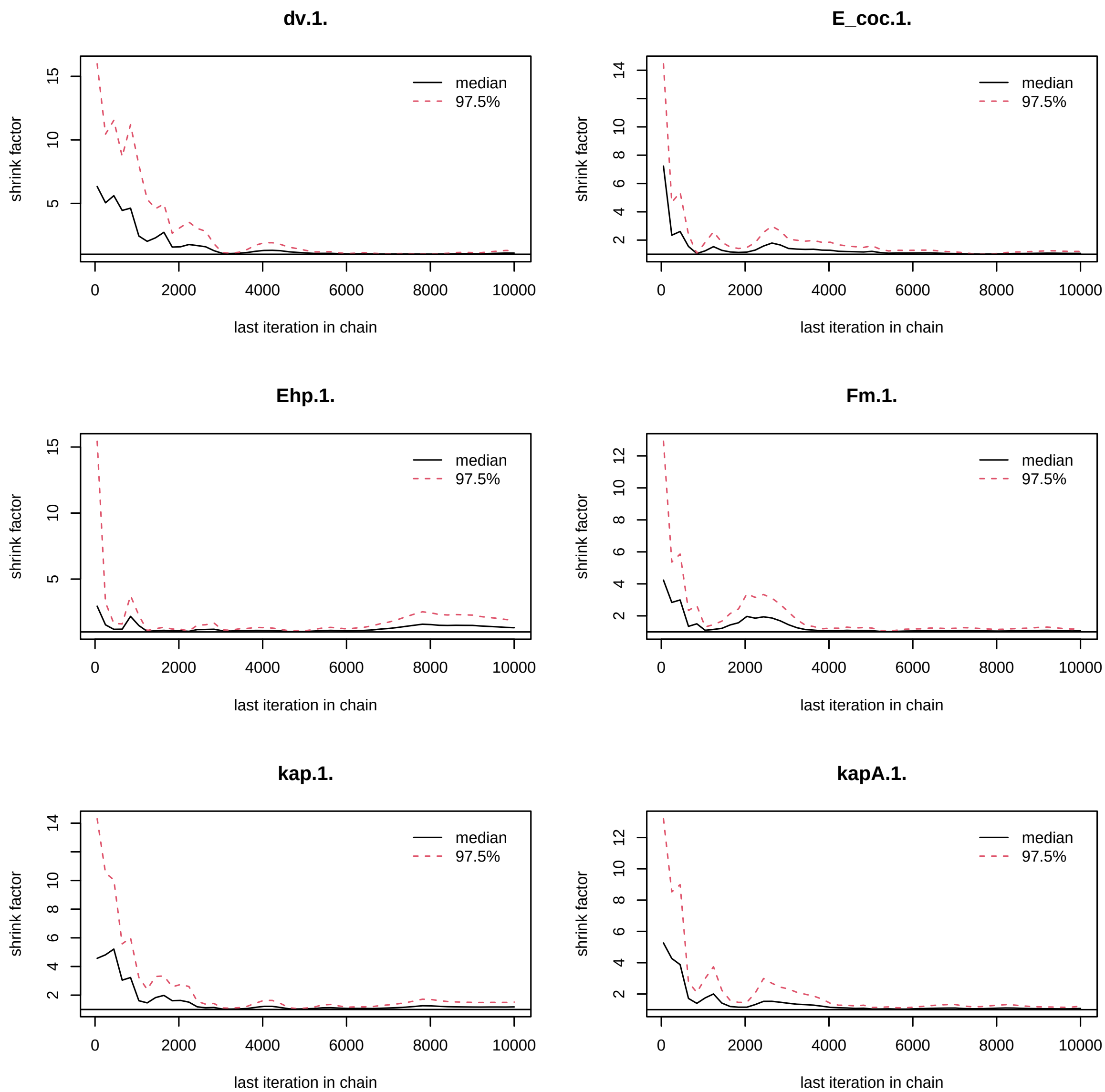


Figure 3: Chains convergence.

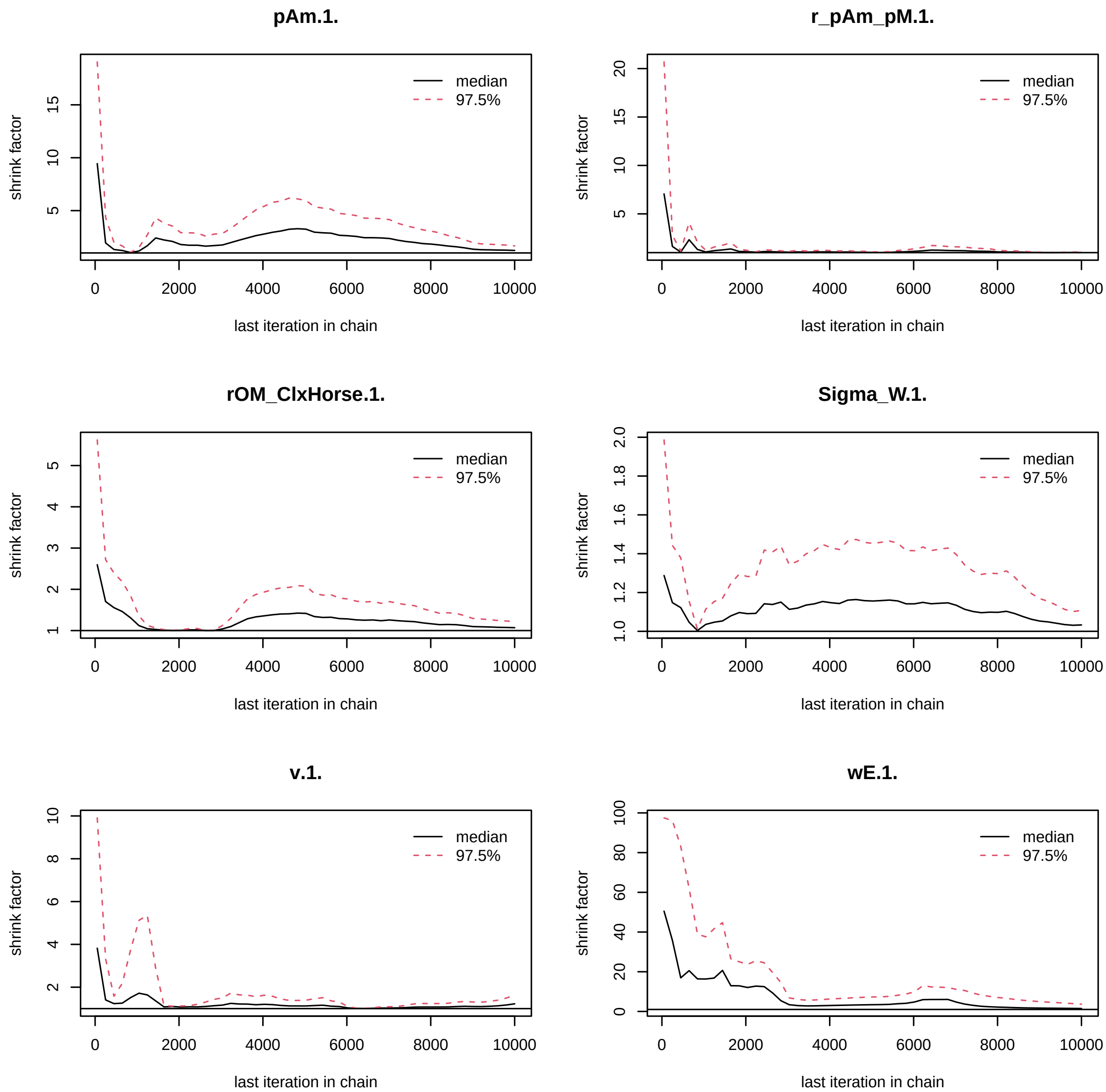


Figure 4: Chains convergence.

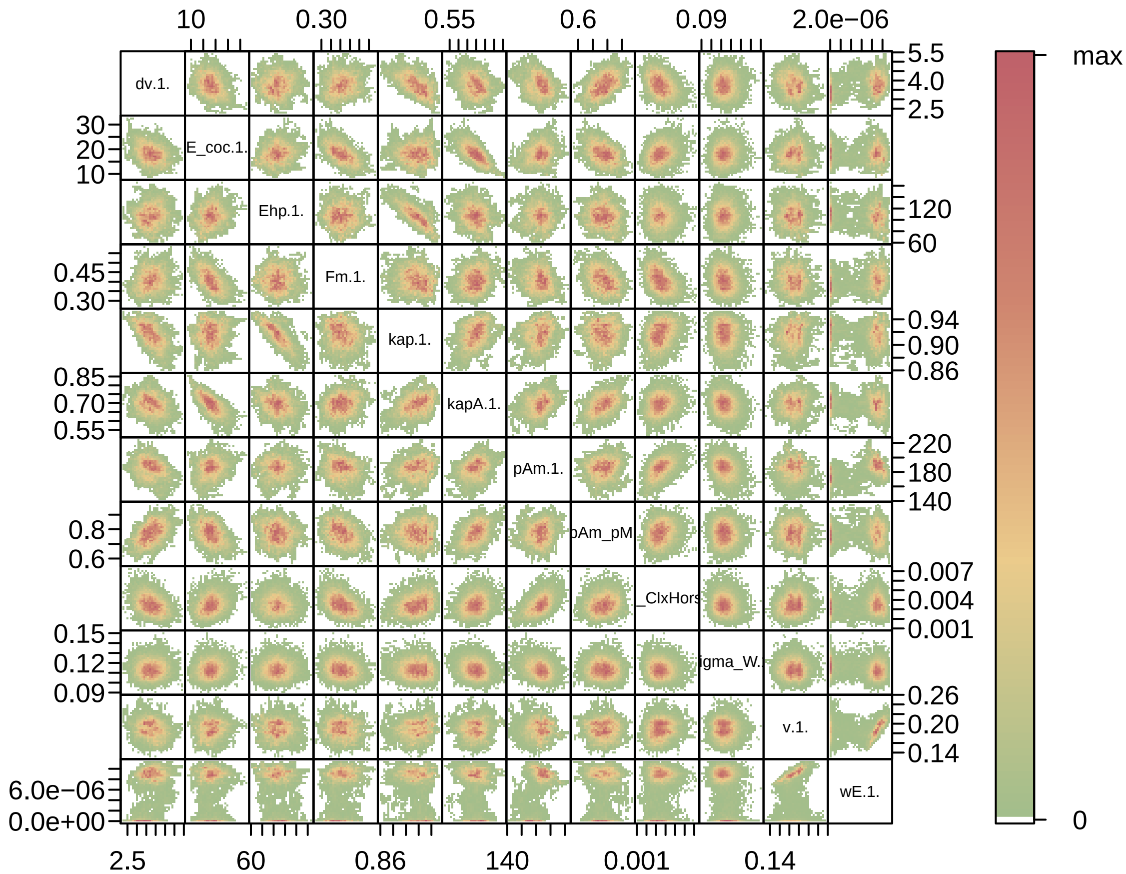


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

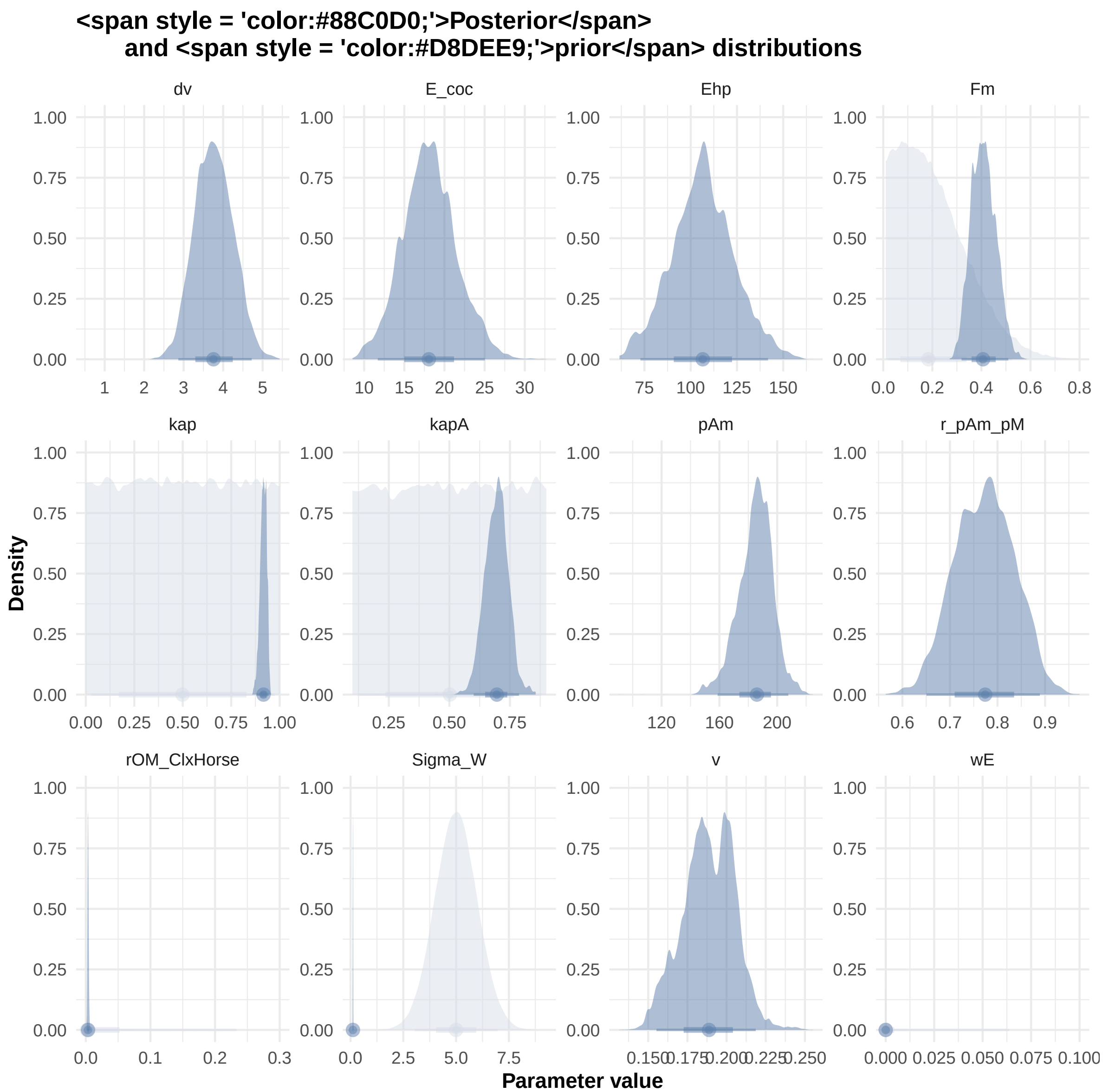


Figure 6: Distributions of the priors and the posteriors.

```
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\.1\\.$|\\.1\\.", "", param),
    line  = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
    Length_struct,
    Reproduction,
    Energy,
    Organic_matter,"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_prin
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr

# ./mcsim.DEBcali DEB_sim_full.in
```

```
##############################
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
```

```r
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#############################

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
```

```
Warning: `read_table2()` was deprecated in readr 2.0.0.
i Please use `read_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpe
i Row 1 of `x` matches multiple rows in `y`.
i Row 1 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```

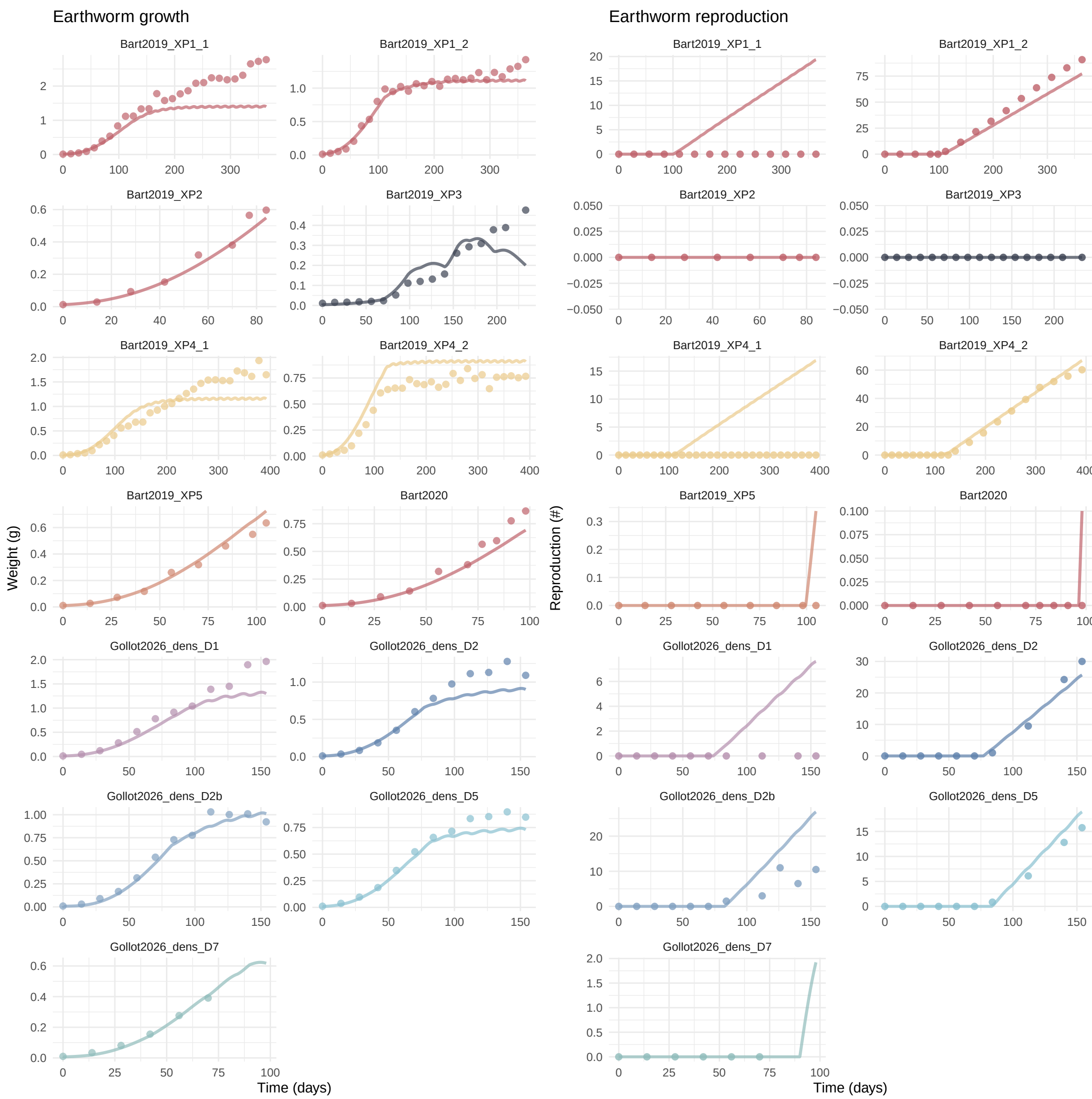
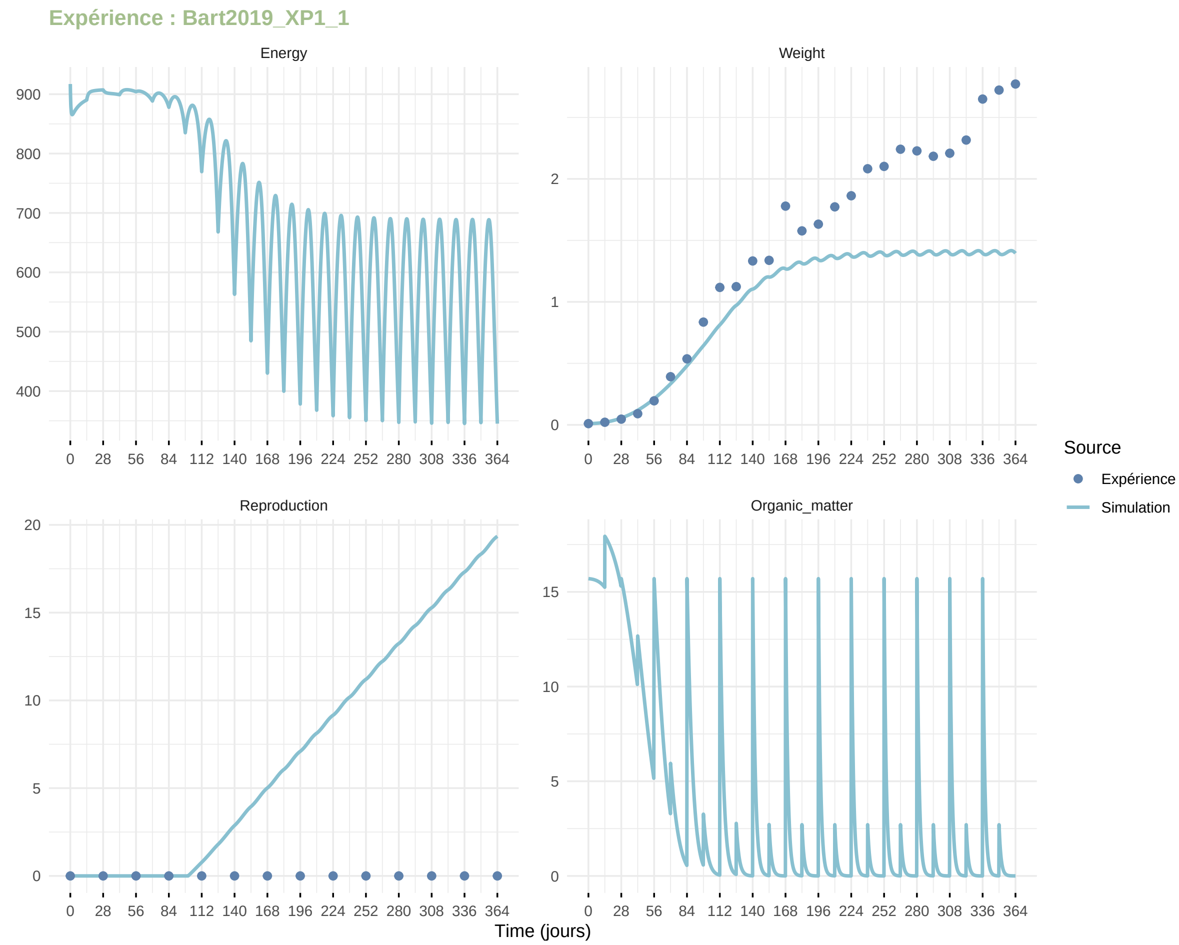
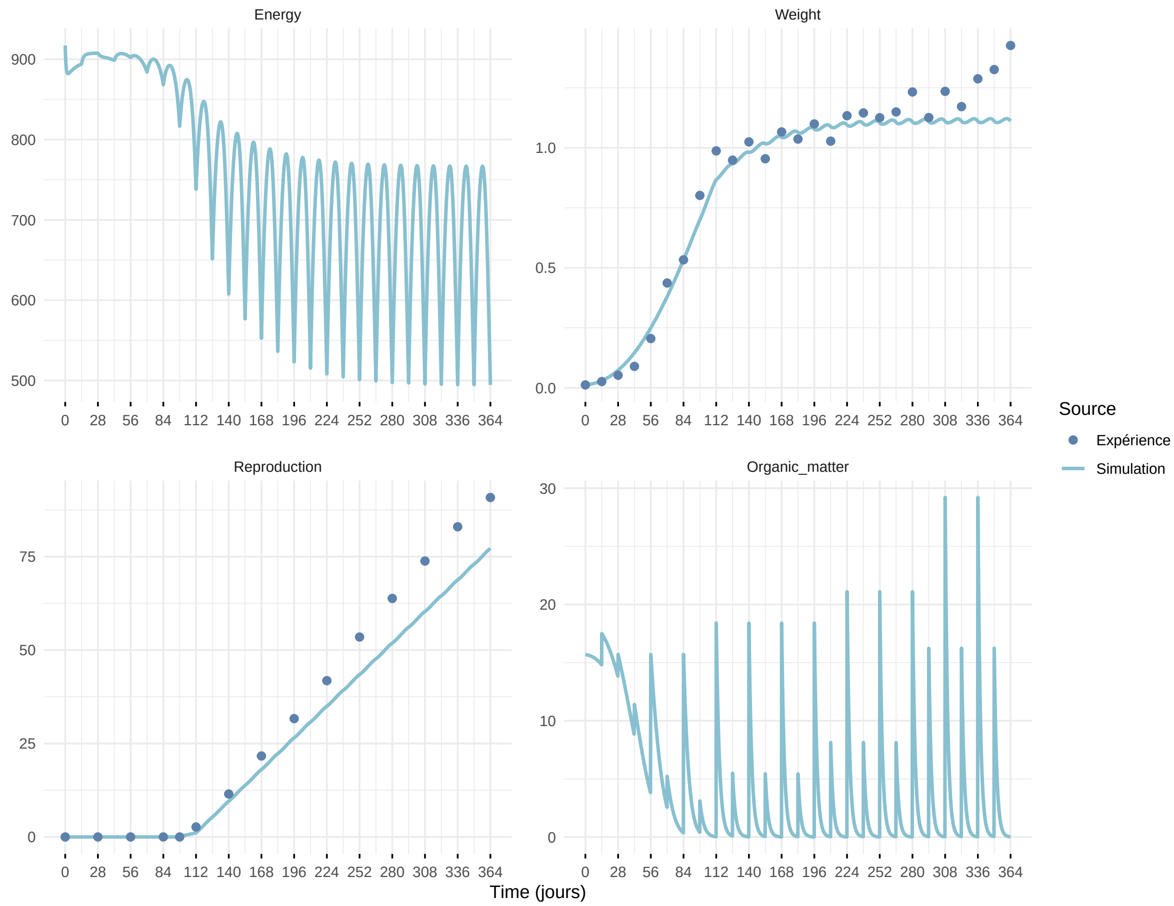
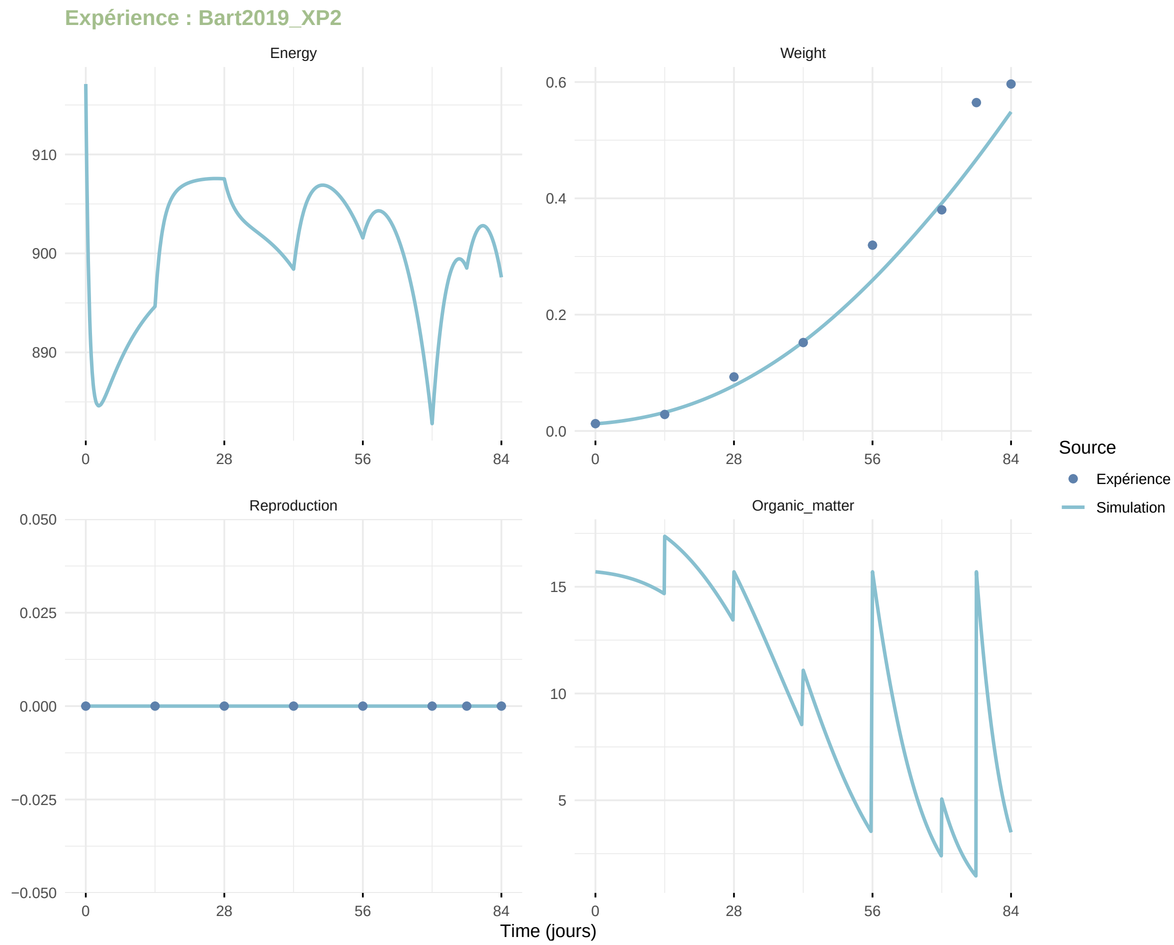


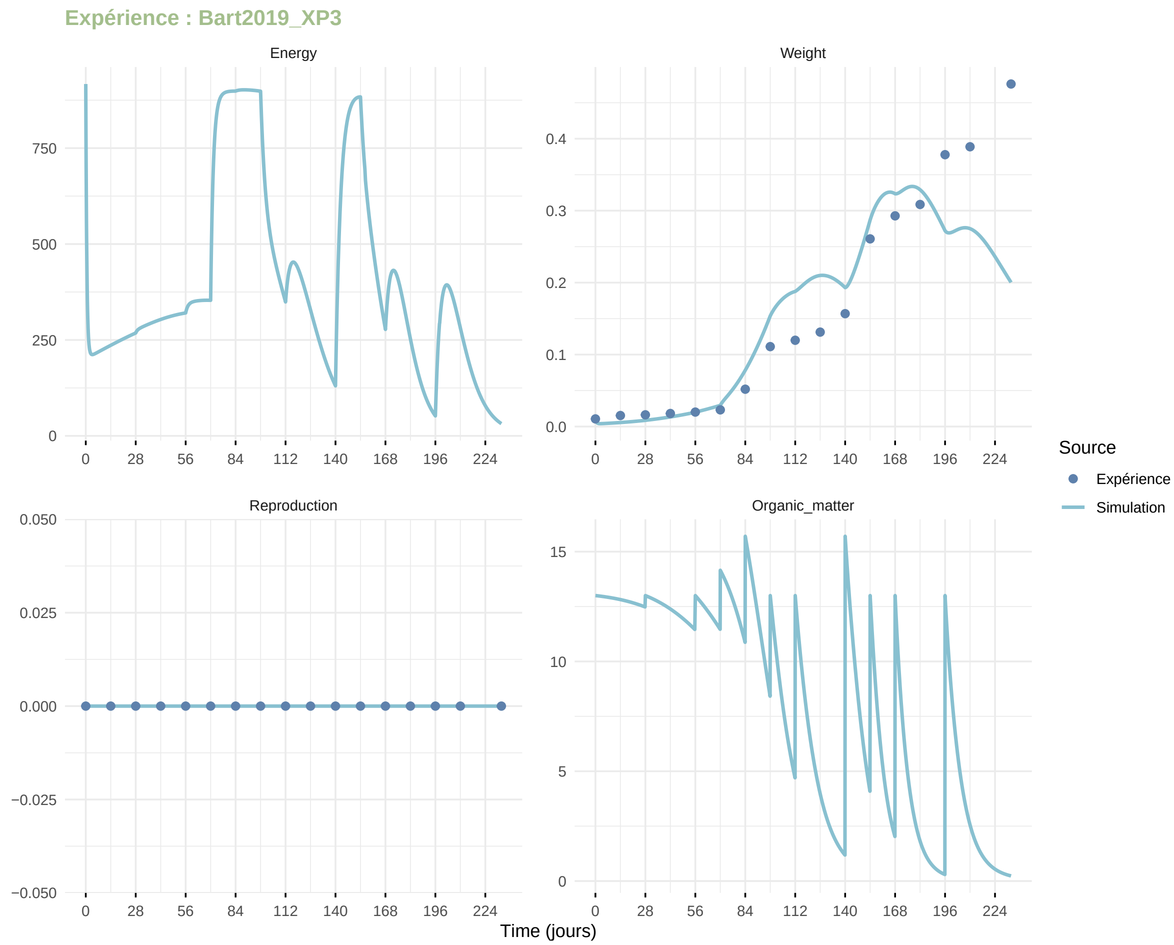
Figure 7: Simulations Weight and Reproduction.


Expérience : Bart2019_XP1_1
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)
0 28 56 84 112 140 168 196 224 252 280 308 336 364

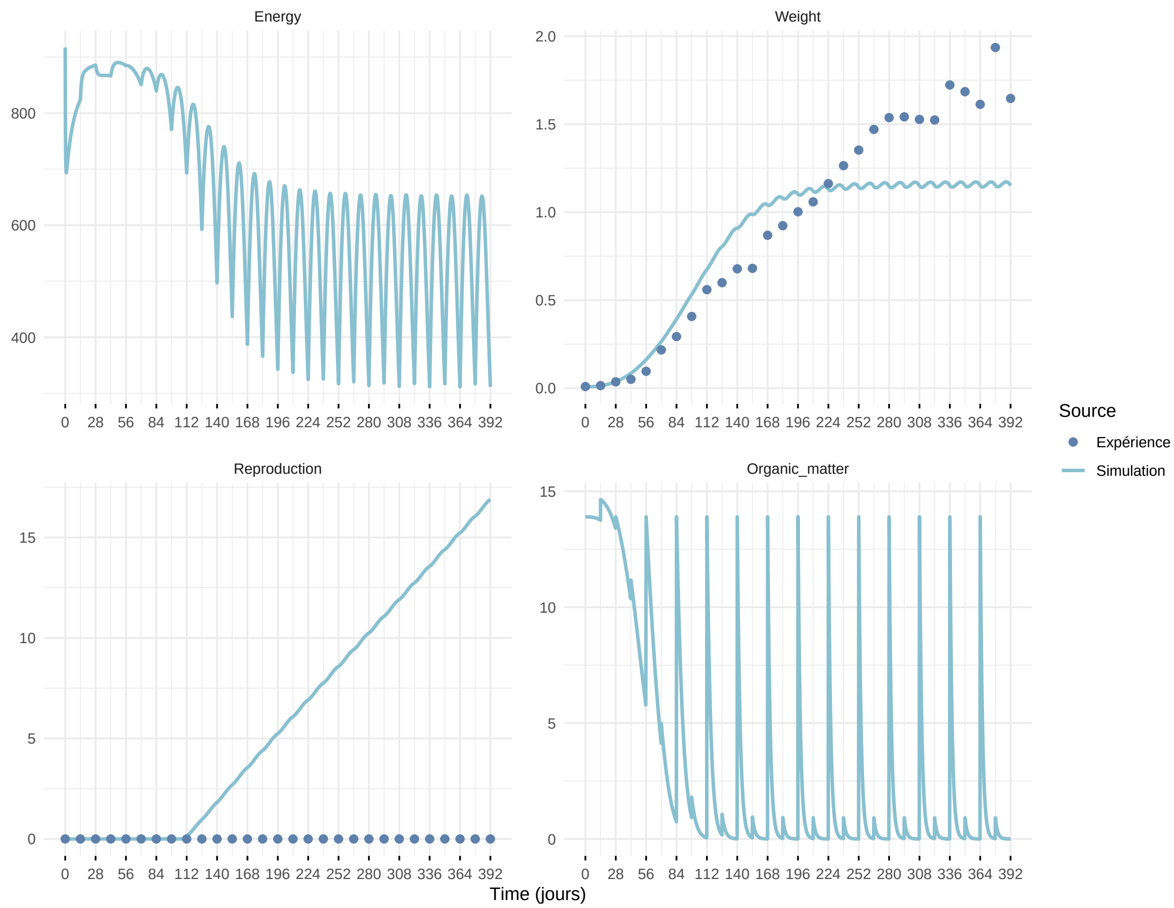
Expérience : Bart2019_XP1_2



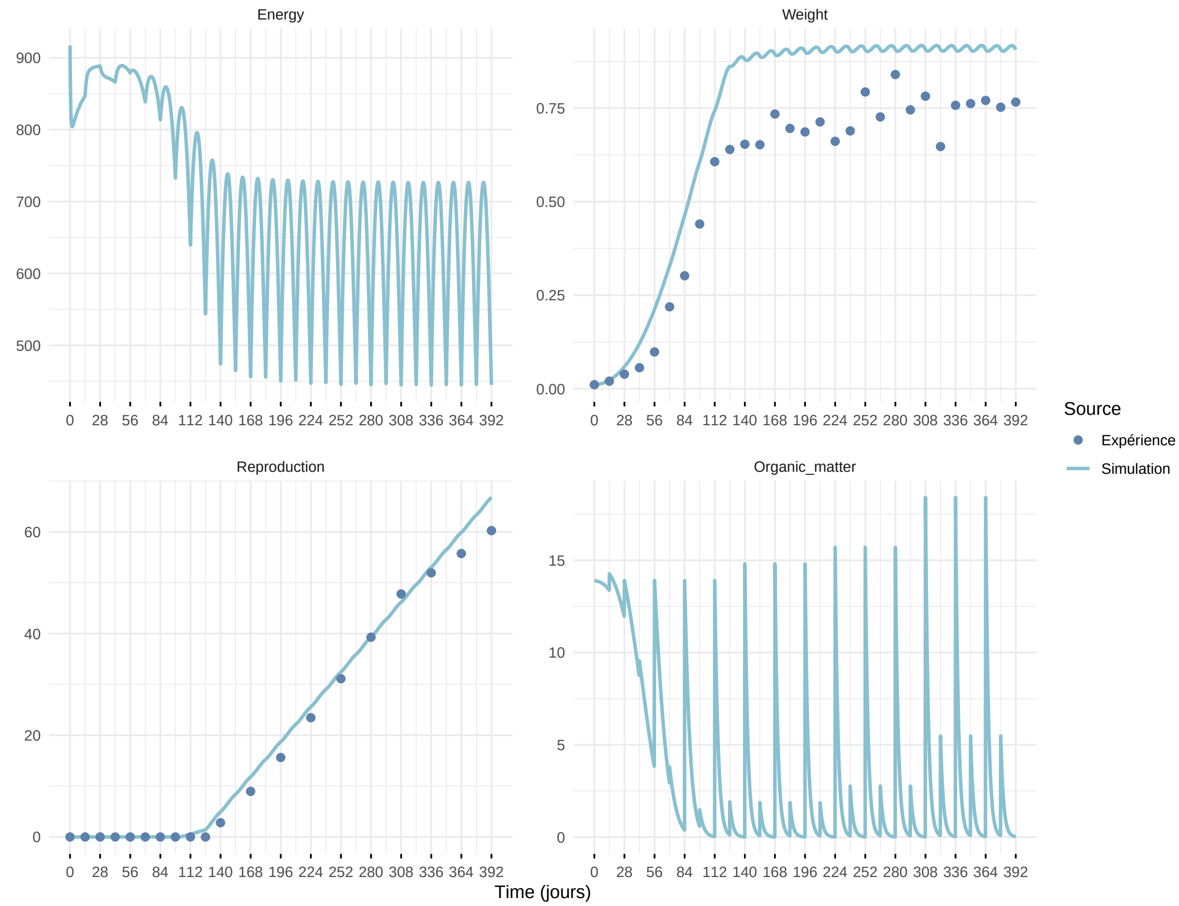

Expérience : Bart2019_XP2
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)
0
28
56
84
910
900
890
0.6
0.4
0.2
0.0
0.050
0.025
0.000
−0.025
−0.050
15
10
5

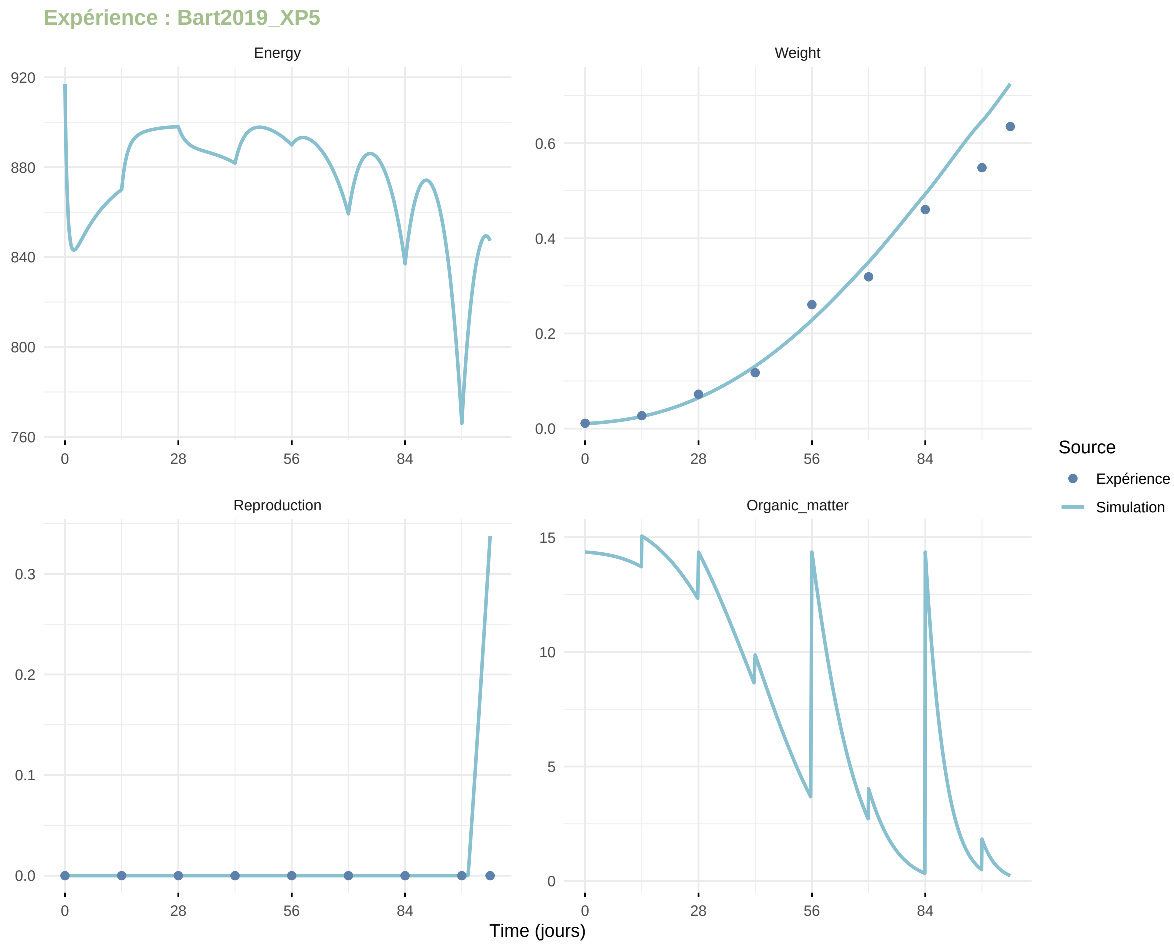

Expérience : Bart2019_XP3
Energy
Weight
Reproduction
Organic_matter
0
250
500
750
0.0
0.1
0.2
0.3
0.4
0.050
0.025
0.000
−0.025
−0.050
0
5
10
15
0
28
56
84
112
140
168
196
224
Time (jours)
Source
Expérience
Simulation

Expérience : Bart2019_XP4_1

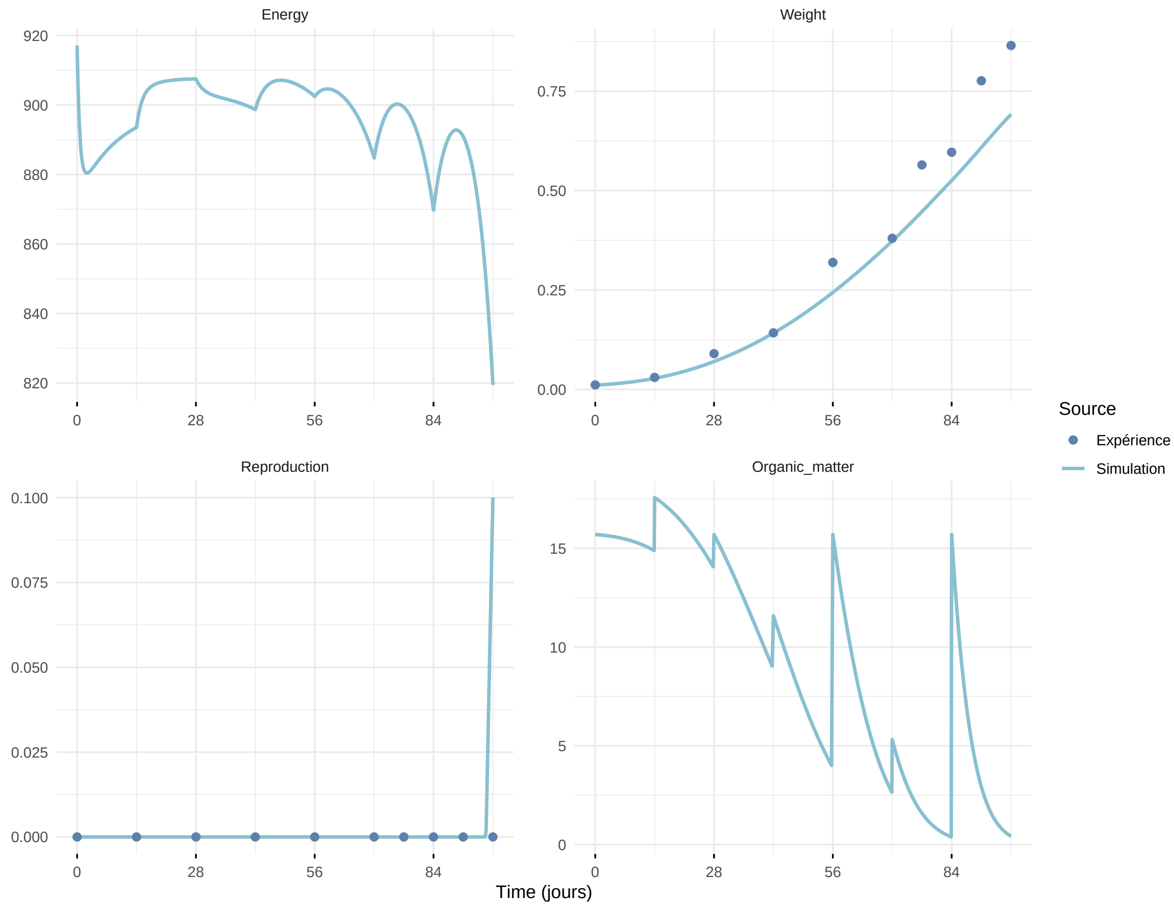


Expérience : Bart2019_XP4_2

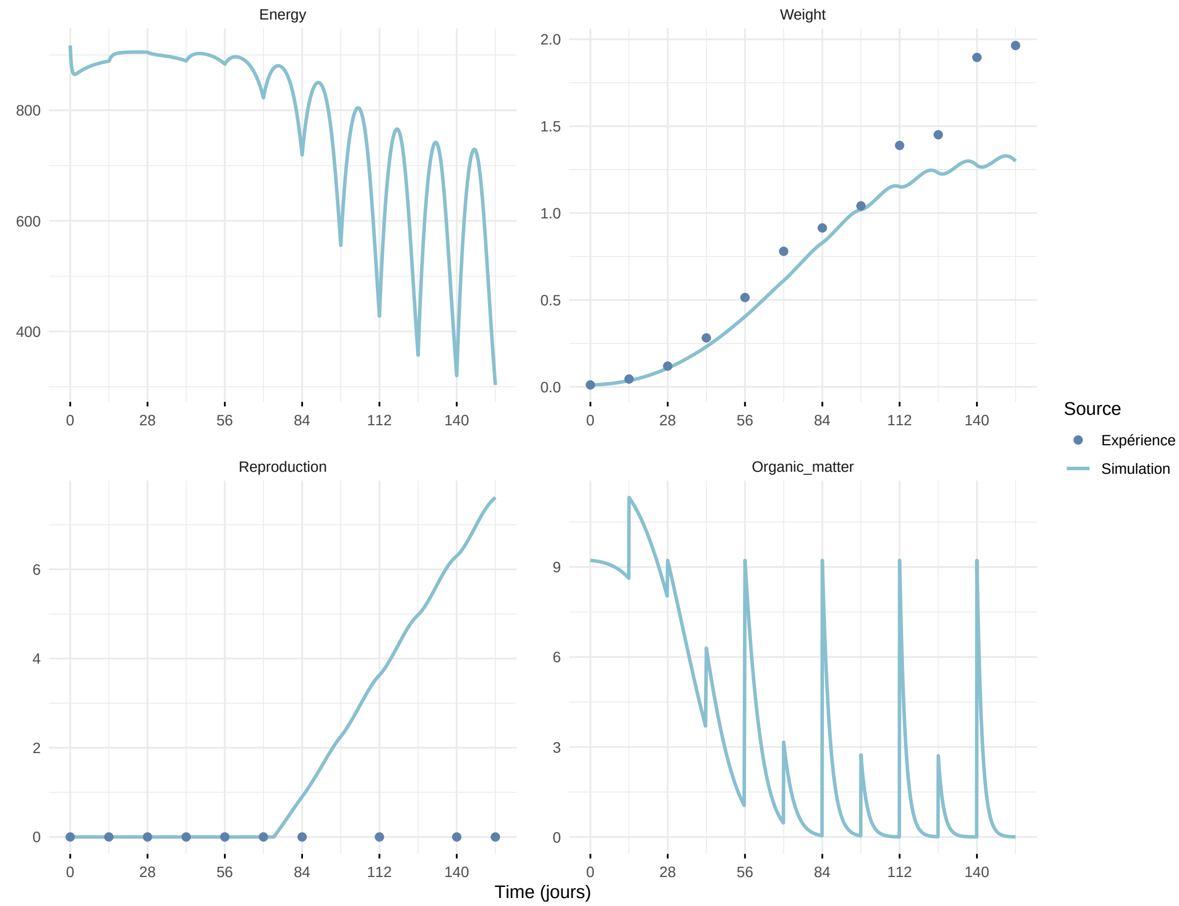



Expérience : Bart2019_XP5
Energy
Weight
Reproduction
Organic_matter
920
880
840
800
760
0.6
0.4
0.2
0.0
0.3
0.2
0.1
0.0
15
10
5
0
0
28
56
84
Source
Expérience
Simulation
Time (jours)

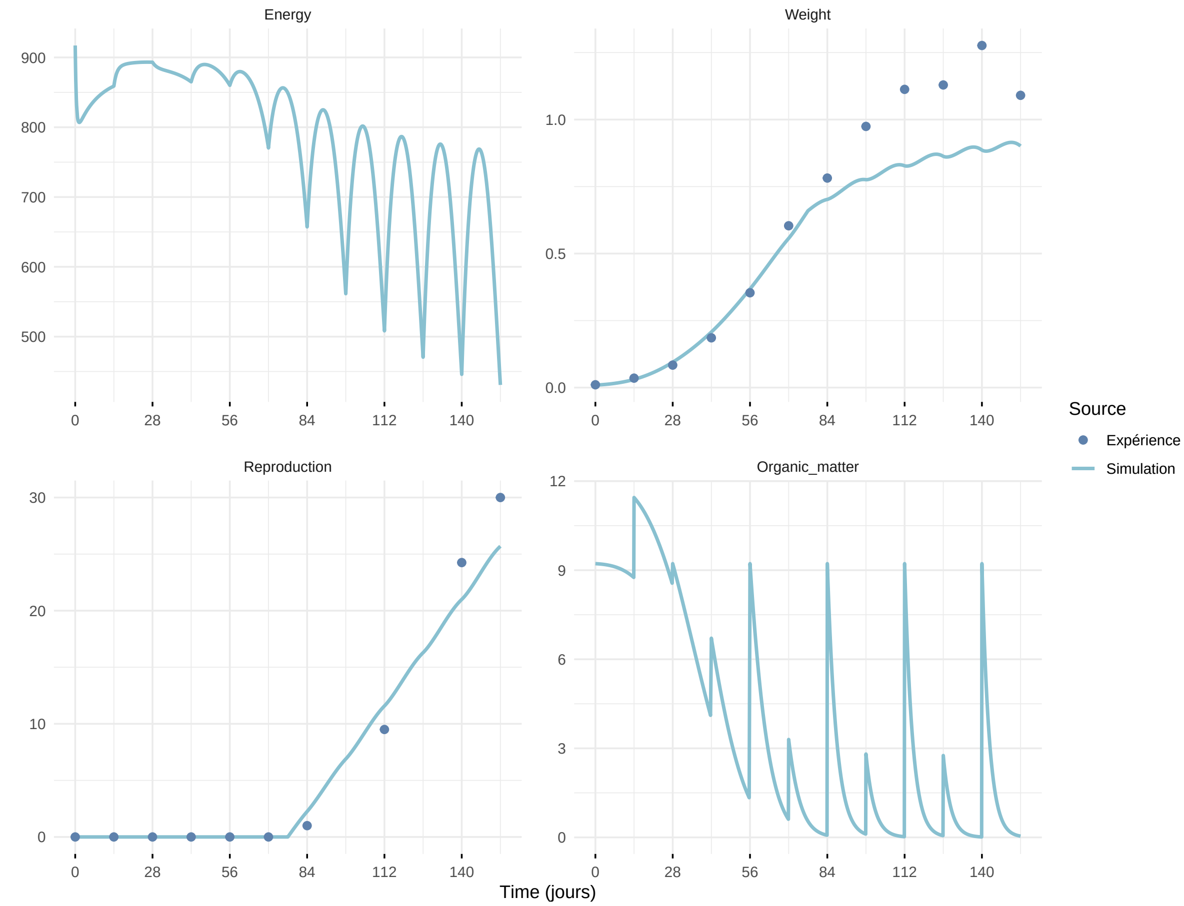
Expérience : Bart2020

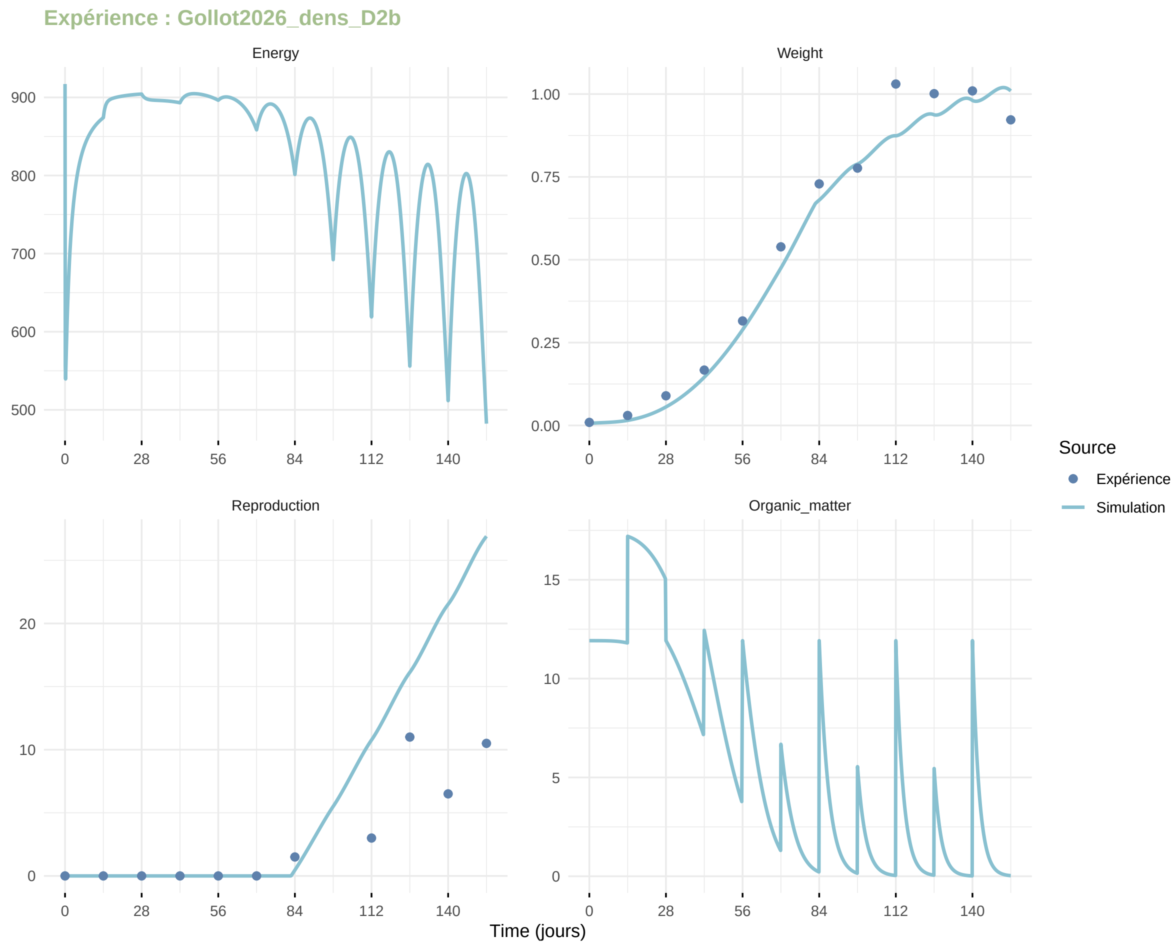


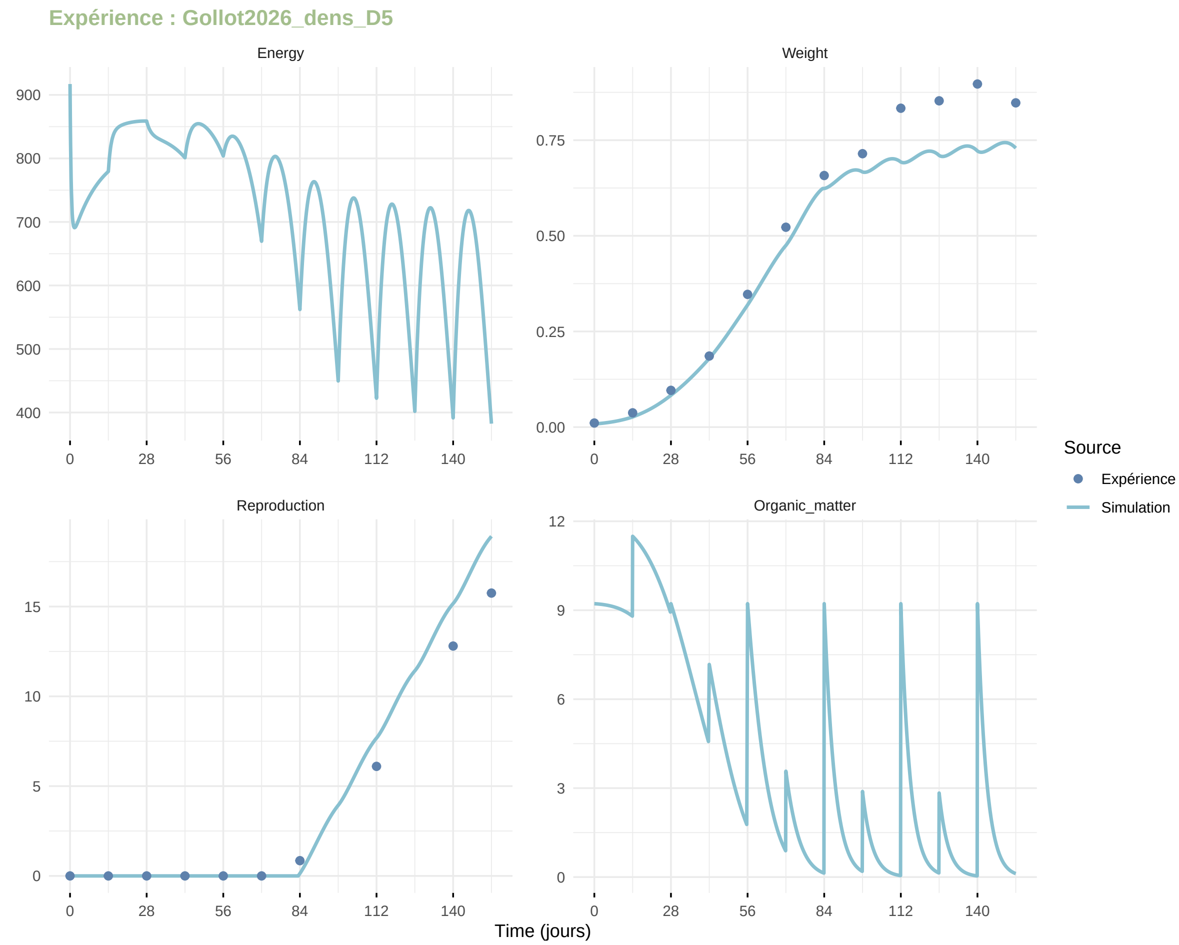
Expérience : Gollot2026_dens_D1



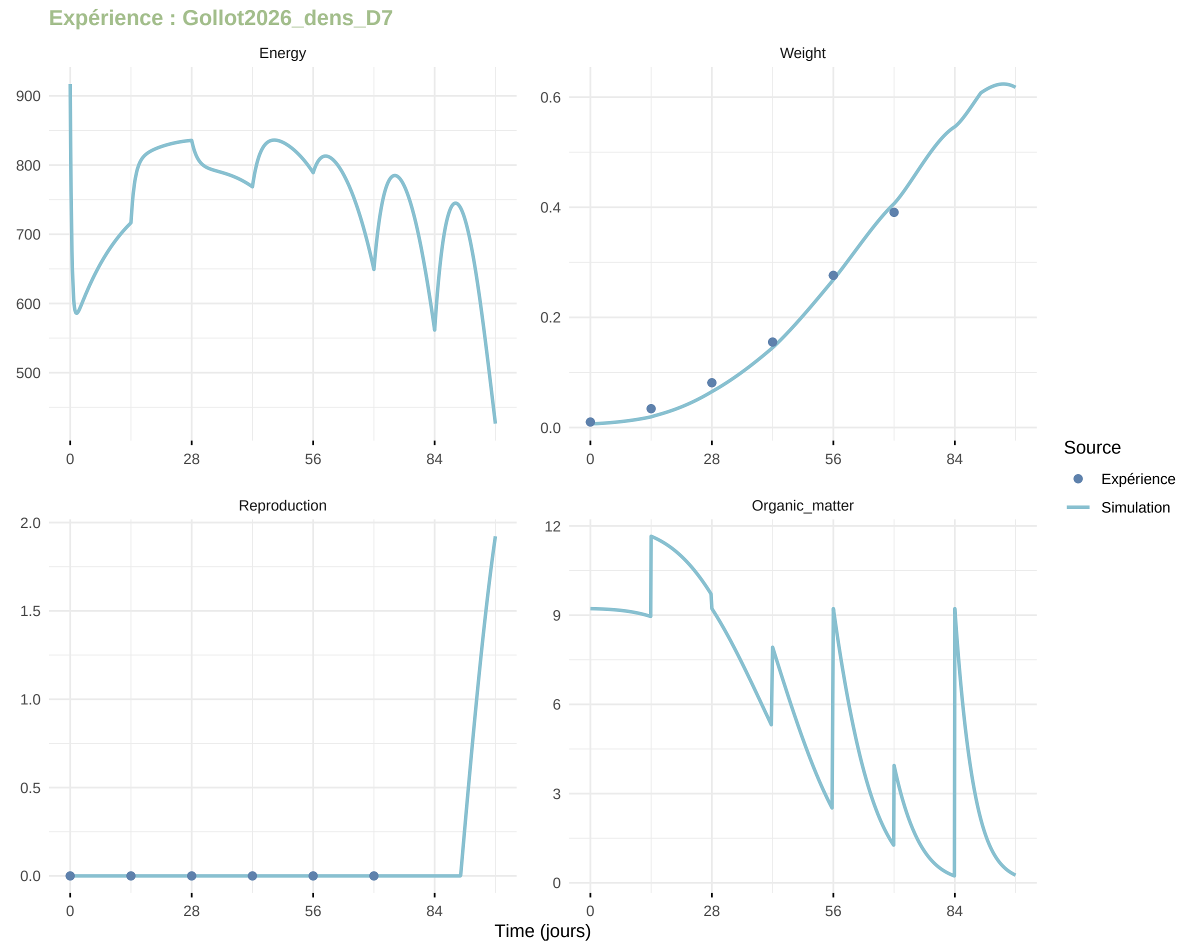
Expérience : Gollot2026_dens_D2




Expérience : Gollot2026_dens_D2b
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)
0
28
56
84
112
140
500
600
700
800
900
0.00
0.25
0.50
0.75
1.00
0
10
20
0
5
10
15



Expérience : Gollot2026_dens_D5
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)


Expérience : Gollot2026_dens_D7
Energy
Weight
Reproduction
Organic_matter
Time (jours)
Source
Expérience
Simulation